

Overview:

This lab focused on wireless network reconnaissance, traffic analysis, rogue access point deployment, and WPA security assessment. The objective was to understand common wireless attack techniques, observe wireless communications, and evaluate the security risks associated with improperly secured wireless networks.

Tools used:

- Kali Linux
- Aircrack-ng Suite
- Airgraph-ng
- Wireshark
- Wireless Network Adapter (Monitor Mode Capable)
- Linux Command Line

Skills Demonstrated:

- Wireless Security Assessment
- Network Traffic Analysis
- Packet Capture and Inspection
- Access Point Enumeration
- Wireless Reconnaissance
- Security Risk Analysis
- Cybersecurity Documentation

1. MAC Address Modifications and Security Implications:

A MAC address uniquely identifies a network interface. By changing or spoofing a MAC address, an attacker can disguise their device as another system. Potential attacks include MAC spoofing, bypassing MAC filtering, evasion of tracking mechanisms, and device impersonation. This demonstrates why MAC filtering should not be relied upon as a primary security control.

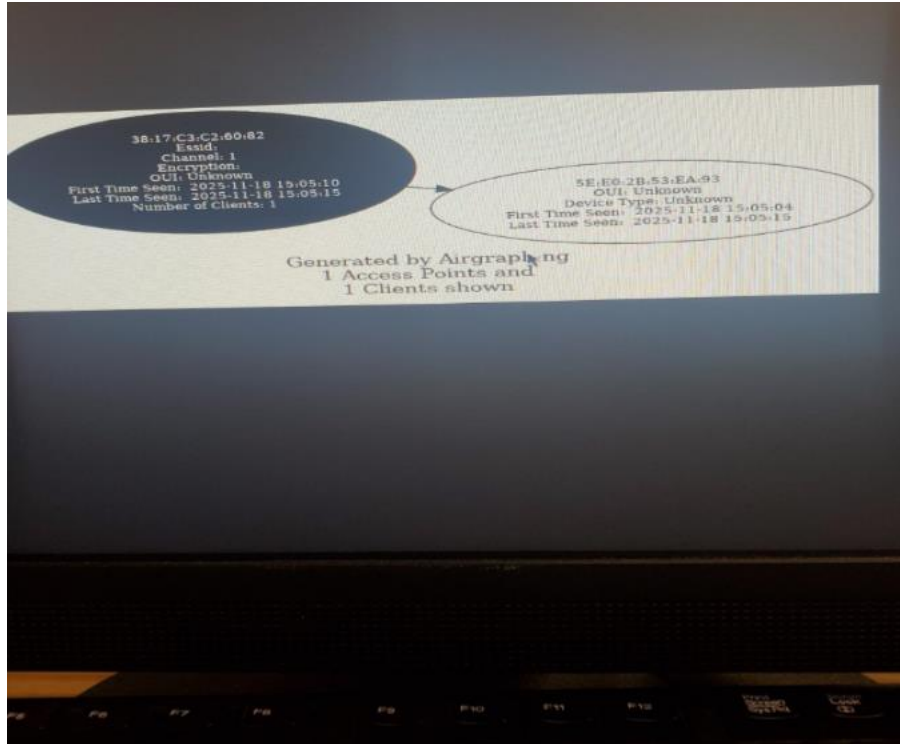
Here is an example that I did showing how to change a mac address.

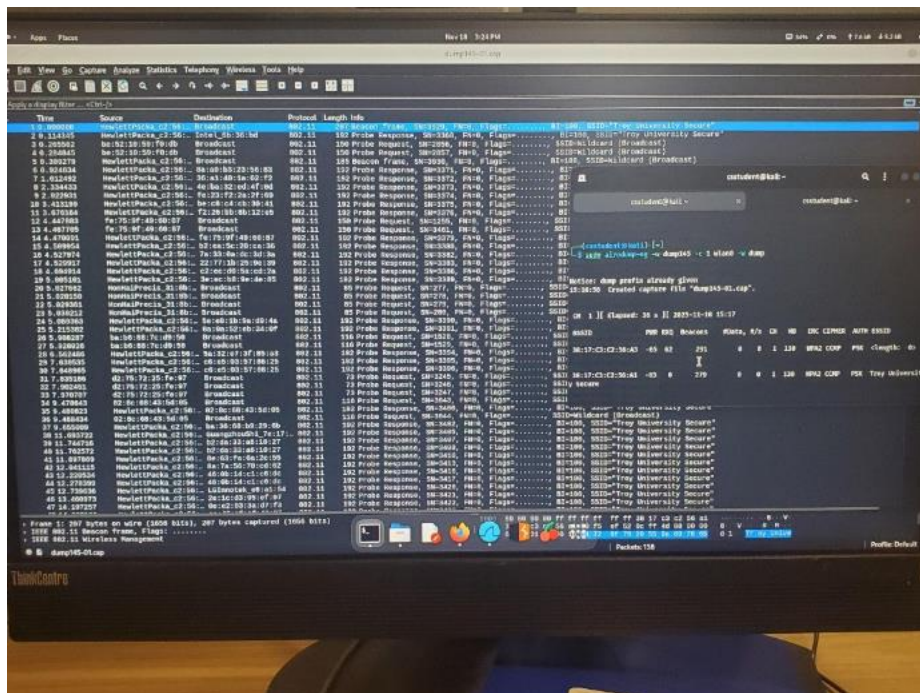
```
csstudent@kali: ~  
[csstudent@kali]~  
$ sudo ifconfig wlan0 down  
[sudo] password for csstudent:  
[csstudent@kali]~  
$ sudo macchanger -r wlan0  
Current MAC: 98:48:27:e8:f3:4b (unknown)  
Permanent MAC: 98:48:27:e8:f3:4b (unknown)  
New MAC: aa:12:6b:67:79:29 (unknown)  
[csstudent@kali]~  
$ sudo ifconfig wlan0 down  
[csstudent@kali]~  
$
```

2. Monitoring Traffic Between Access Points and Stations:

Wireless traffic was captured between access points and client stations. Airgraph-ng was used to visualize relationships between devices, while Wireshark was used to analyze management and data frames. Observations included beacon frames, probe requests, authentication frames, and device manufacturer identification through OUI lookups. Monitoring channels 1, 6, and 11 provided insight into network activity and utilization.

Here are pictures of my work.





3. Evil Twin Access Point Attack:

An Evil Twin attack involves creating a rogue access point that mimics a legitimate wireless network. Users may unknowingly connect to the rogue network. This attack can facilitate Man-in-the-Middle (MITM) attacks, credential harvesting, and traffic monitoring. The exercise highlights the importance of verifying network authenticity and using encrypted protocols.

Here is an example of my work.

```
Session Actions Edit View Help
(csstudent@kali)-[~]
└─$ sudo airmon-ng start wlan0
[sudo] password for csstudent:

PHY      Interface      Driver      Chipset
phy1     wlan0           rtlw88_8814au  TP-Link Archer T9UH v1 [Realtek RTL8814AU]
          (mac80211 monitor mode already enabled for [phy1]wlan0 on [phy1]10)

(csstudent@kali)-[~]
└─$ sudo airbase-ng -e IamVenom -c 6 wlan0
16:19:24 Created tap interface at0
16:19:24 Trying to set MTU on at0 to 1500
16:19:24 Access Point with BSSID 46:A7:88:83:89:14 started.
16:25:03 Client 12:89:19:44:AD:33 associated (unencrypted) to ESSID: "IamVenom"
16:25:03 Client 12:89:19:44:AD:33 associated (unencrypted) to ESSID: "IamVenom"
16:25:03 Client 12:89:19:44:AD:33 associated (unencrypted) to ESSID: "IamVenom"
16:25:03 Client 12:89:19:44:AD:33 associated (unencrypted) to ESSID: "IamVenom"
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16:25:03 Client 12:89:19:44:AD:33 associated (unencrypted) to ESSID: "IamVenom"
```

4. WPA Security Assessment and Password Recovery Testing:

This assessment examined WPA security by capturing authentication material and evaluating password strength through offline password recovery testing. Results demonstrated that weak passwords are significantly more susceptible to dictionary attacks, while strong passphrases greatly improve resistance against password recovery attempts.

Picture of my work.

```
Aircrack-ng 1.7
[00:00:00] 24/14344392 keys tested (948.69 k/s)
Time left: 4 hours, 12 minutes, 0 seconds          0.00%
KEY FOUND! [ 123456789 ]

Master Key      : 33 1D 0E FD 03 D3 4C E2 CA B7 6F 38 CB 28 E4 75
                  57 48 47 38 6B 06 09 2B 22 AC 5F 1C CA 67 E2 D5
Transient Key    : C0 0B C7 16 93 FA D5 D5 27 99 A1 B8 55 D0 26 3A
                  99 19 9D 42 27 98 BE E2 1E 98 76 90 AB FF E3 93
                  C7 6F 64 B0 41 4A BC 50 00 D5 0B CC 9C 9A 14 51
                  BB 5D 06 21 69 9E ED 11 80 4C D7 0C FC 13 9E 00
EAPOL HMAC      : BC 03 BA BF A5 8F 72 6C D9 A7 B5 1C A5 1A E1 42

(csstudent@kali)I[-]
```

What I learned:

Wireless networks expose significant information through management of traffic. Strong authentication, WPA2/WPA3 security, complex passwords, and continuous monitoring are critical defenses. This lab reinforced the importance of layered security controls and user awareness.

Conclusion:

This project provided practical experience with wireless reconnaissance, packet analysis, rogue access points, and wireless authentication security. The assessment demonstrated both common attack techniques and defensive strategies used to secure modern wireless environments.

